

# Building a Telugu + Indian-English TTS Dataset

Sarvam AI take-home

## The short version

I set out to build a 60-minute, single-speaker text-to-speech dataset – roughly half Telugu, half Indian English – where every clip is one person speaking cleanly, with an accurate transcript and a plain-English sentence describing *how* it's said (the speaker's emotion, style, pace, pitch). I collected a lot more than I needed from 14 YouTube sources, ran it through a quality pipeline, listened to the data (my grandmother helped with the Telugu), and trimmed it down to a balanced set.

**Where it landed:** 190 clips, 79 minutes, split evenly (95 Telugu / 95 English). This is a deliberately over-sized "review buffer" – bigger than 60 minutes on purpose, so clips can be thrown out during review and the set still clears the 30-min-per-language bar. The script `pipeline/balance.py` re-trims it to the final size whenever I want.

The assignment says it's graded on *data quality and judgment, not code*, so most of this report is the story: what I built, the things that went wrong, and what I'd fix.

## A quick glossary

A few terms show up throughout. In plain language:

| Term                            | What it means here                                                                                                                               |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ASR</b>                      | Automatic Speech Recognition – turning audio into text. Sarvam's model is <code>saaras:v3</code> .                                               |
| <b>VAD</b>                      | Voice Activity Detection – finding where speech is vs. silence, so I can cut clips at pauses.                                                    |
| <b>Diarization</b>              | "Who spoke when" – splitting a recording by speaker, so I can keep just one person.                                                              |
| <b>SER</b>                      | Speech Emotion Recognition – a model that guesses emotion from the <i>sound</i> , not the words.                                                 |
| <b>Parler-style description</b> | A natural-language sentence describing delivery (e.g. "A woman narrates slowly in a calm voice"), the format the Parler-TTS project popularised. |

| Term                  | What it means here                                                                                                 |
|-----------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Code-switching</b> | Mixing two languages in one sentence – extremely common in Indian speech (Telugu with English words).              |
| <b>WER / CER</b>      | Word / Character Error Rate – how far a transcript is from the correct one (lower is better).                      |
| <b>Manifest</b>       | My single bookkeeping file, <code>data/manifest.jsonl</code> – one row per clip holding everything known about it. |

## The sources I collected (what the names mean)

The dataset is only as good as what goes in, so I picked sources for *variety* – different speakers, genders, and speaking styles – not just whatever was easy. Each gets a short code-name in the config:

| Source name     | What it actually is                                                  | Lang | Why I chose it                              |
|-----------------|----------------------------------------------------------------------|------|---------------------------------------------|
| Harshaneeyam    | A Telugu literary <b>storytelling</b> podcast                        | te   | emotional range (sad, excited)              |
| BV_Pattabhiram  | A Telugu <b>motivational / counselling</b> lecture                   | te   | calm, serious delivery                      |
| Garikapati      | Garikapati N. Rao's Telugu <b>literary discourse</b>                 | te   | oratorical, animated                        |
| Te_Standup_M    | A Telugu <b>stand-up comedy</b> set (male)                           | te   | playful/happy + a crowd-noise test          |
| Te_ASMR         | Telugu <b>ASMR</b> (whispered)                                       | te   | the only way to get <i>whisper</i> clips    |
| En_Podcast_Raj  | A long <b>podcast</b> that turned out to be Telugu (see iteration 6) | te   | conversational; added a female Telugu voice |
| TED_India       | A <b>TEDx India</b> talk                                             | en   | clear oratorical English                    |
| EN_Audiobook    | An Indian-English <b>audiobook</b> narration (female)                | en   | clean narrative reference                   |
| EN_Podcast_Solo | A solo Indian-English <b>podcast</b> monologue (male)                | en   | conversational                              |

| Source name         | What it actually is                                                    | Lang | Why I chose it                         |
|---------------------|------------------------------------------------------------------------|------|----------------------------------------|
| En_Comedy           | "A Moment of Silence", a <b>multi-host comedy podcast</b> (female-led) | en   | playful/happy; diarized to one speaker |
| En_Speech           | An Indian-English <b>speech</b>                                        | en   | oratorical                             |
| En_Standup_F1/F2/F3 | Three <b>female stand-up</b> specials                                  | en   | fix a male-heavy roster; happy/excited |

So when I say "En\_Comedy", that's a real comedy podcast with several hosts talking over each other – which is exactly why it needed diarization to pull out a single voice.

## 1. What I built and how the pipeline works

The whole thing is a chain of small scripts, each doing one job and writing its result back to one file (`data/manifest.jsonl`). Because every clip's progress is recorded in that file, I can stop and restart anywhere – which mattered, because I hit crashes and ran out of API credits more than once, and every restart just picked up where it left off.

Walking through the stages:

- **Download** (s1) – grab each source's audio with `yt-dlp`, convert to 16 kHz mono and even out the loudness with `ffmpeg`.
- **Segment** (s2) – cut each recording into ~25-second clips, but *only at silences*, so a word is never chopped in half. Every clip stays under the 30-second limit Sarvam's ASR needs. For sources with more than one speaker, `s2b` instead runs Sarvam's diarization and keeps only the person who talks the most.
- **Filter** (s3) – listen for problems and *throw clips out*: background music, crowd noise, or two people in one clip. I never try to "repair" audio – for TTS, a cleaned-up clip is worse than no clip.
- **Features** (s4, s4b) – measure pitch, energy, and pace; and separately, run a model that guesses emotion straight from the *sound*.
- **Transcribe** (s5) – send each clip to Sarvam `saaras:v3`, telling it the language up front for accuracy.
- **Tag & describe** (s6, s7) – ask Sarvam's `sarvam-30b` for the emotion, style, and whisper flag, then for the one-sentence description.
- **Export** (s8) – package it as a HuggingFace dataset and publish.

### The one rule I'm proud of

The machine's guesses and the human's corrections are kept in **separate columns that never overwrite each other**. The transcript Sarvam produced lives in `asr_transcript`; if a human fixes it, that goes in `human_transcript`. They're combined only at the very end:

```
final_transcript = human_transcript or asr_transcript  # human wins if present
final_emotion    = human_emotion    or llm_emotion
```

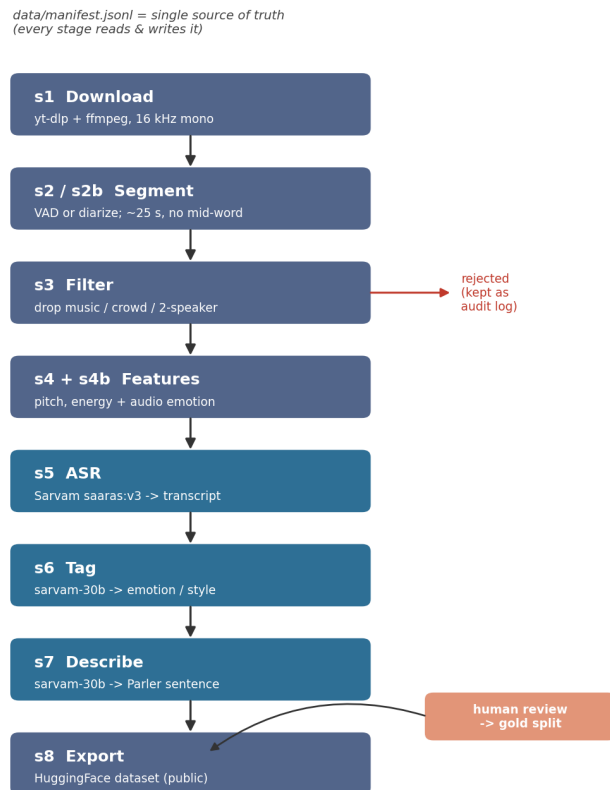


Figure 1: The pipeline. Each stage reads the manifest, does its one job, and writes back. Teal stages call a Sarvam API; the rest are local helpers. Rejected clips are kept (as an audit log), and a human-review step produces the trusted “gold” clips before export.

This means the raw machine output is never lost, so I can always measure honestly how good the ASR was. Human-checked clips become the trusted **gold** set. A manifest row, grouped by who owns each field:

```
identity : clip_id, source_channel, language, gender, duration, ...
machine  : asr_transcript / llm_emotion / llm_style / whisper / audio_emotion ...
human    : human_transcript / human_emotion / human_verified / reviewer ...
```

## 2. Iterations I did to improve data quality

Almost every quality gain came from catching the pipeline quietly doing the wrong thing. In order:

1. **The music filter wasn't filtering anything.** The library I used (`inaSpeechSegmenter`) crashed on every single clip due to a version mismatch – but the crash was swallowed, so the stage “succeeded” while detecting zero music. I had to pin an older Keras and patch a NumPy call. *Lesson: a stage that fails silently is more dangerous than one that crashes loudly.*
2. **70 good clips were wrongly thrown out as “two speakers.”** The gender detector was labelling chunks of one expressive male storyteller as “female” (his pitch swings fooled it), which tripped the two-speaker check. I added a per-source `solo: true` flag meaning “I listened, it's one person – skip that check.”
3. **The LLM returned blank answers and burned through credits.** `sarvam-30b` is a *reasoning* model: left alone it spent its entire token budget “thinking” and returned nothing. Every call was both expensive and useless. The fix was one setting (`reasoning_effort=None`) that turns thinking off – about 60x cheaper and 30x faster.
4. **A whole source was silently skipped** because its YouTube id had an underscore in it, which broke my filename parsing. Fixed by matching against the known source names.
5. **Clips started too abruptly.** I was cutting flush to the first sound, which clipped the start of words. Added a small 120 ms cushion taken from the surrounding silence. For ASMR I also had to stop long silent gaps from being packed into clips.
6. **A source I labelled “English” was actually Telugu.** `En_Podcast_Raj` is Telugu with heavy English mixing, but I'd set it to English – so Sarvam faithfully wrote the Telugu in Latin letters (“*O pakka divorce teesukunna vaadu...*”). It *sounded* Telugu but *read* English. Only listening caught it. Relabelling it Telugu and re-running gave proper Telugu script, and I noticed the main speaker is female – which helped balance the dataset.
7. **Whisper can't be detected from text.** The LLM only sees the words, so it can't know a clip is whispered. For ASMR sources I just tell it directly with a `whisper: true` flag.
8. **Over-collect, then balance.** I gathered 617 clips, then wrote `balance.py` to trim down to an even spread across every source and both languages, always keeping any clip a human had already checked.

## 3. What worked and what didn't

### Worked

- Cutting only at silences: all 190 clips land between 18 and 28 seconds, none over the limit, no chopped words.
- Sarvam's ASR is strong on clean audio – on the clips a human checked, the transcripts were good enough to accept as-is. It also handles Telugu-English mixing and writes the English in Telugu script.
- Diarization cleanly pulled single speakers out of 3-4-person podcasts.
- The crowd-noise filter removed 79 noisy stand-up clips.
- The diversity push worked: I went from an almost entirely “narrative + neutral” set to one with real conversational and playful clips, and from 90% male to ~40% female.

### Didn't

- **Some crowd noise still slips through.** My noise check is a blunt “is more than 25% of the clip noisy?” rule, and the detector can't tell “clean voice” from “voice with light laughter underneath”, so a few stand-up clips kept faint audience noise. This is the dataset's weakest spot.
- **English ASMR produced zero clips** – it's whispered so quietly that my silence detector treated the whole thing as silence.
- The audio-emotion model and the text-LLM only agree ~15% of the time on Telugu, because that model was trained on English/German. I use it only to *flag* clips for human attention, never as a label.
- The set still leans narrative – long-form Indian content is mostly monologue.

## 4. Quality observations and decisions I made

The final set, by the numbers:

|                   |                                                                      |
|-------------------|----------------------------------------------------------------------|
| Language          | Telugu 39.7 min (95 clips) · English 39.2 min (95)                   |
| Gender            | male 41 min · female 31 min (~40%) · unknown 7                       |
| Whisper clips     | 5                                                                    |
| Collected -> kept | 617 -> 190 (dropped: 60 music, 79 crowd, 13 by hand, 275 to balance) |

**On accuracy (WER/CER).** Measured on the 37 human-checked clips, error rate is 0.00 – but I want to be honest about what that means: the reviewers *accepted Sarvam's transcripts unchanged*, so this is really an “approval rate” on clean clips, not a strict error rate against a from-scratch transcription. It's a genuinely good sign for Sarvam on clean audio, but a stricter test (re-transcribing blind) would give a truer number.

### Decisions worth calling out:

- **Two opinions on emotion.** The text LLM reads the words; a separate model listens to the sound. When they disagree (they did on 161 of 190 clips), that clip goes to the front of the human-review queue – that's where a human's time is worth the most.

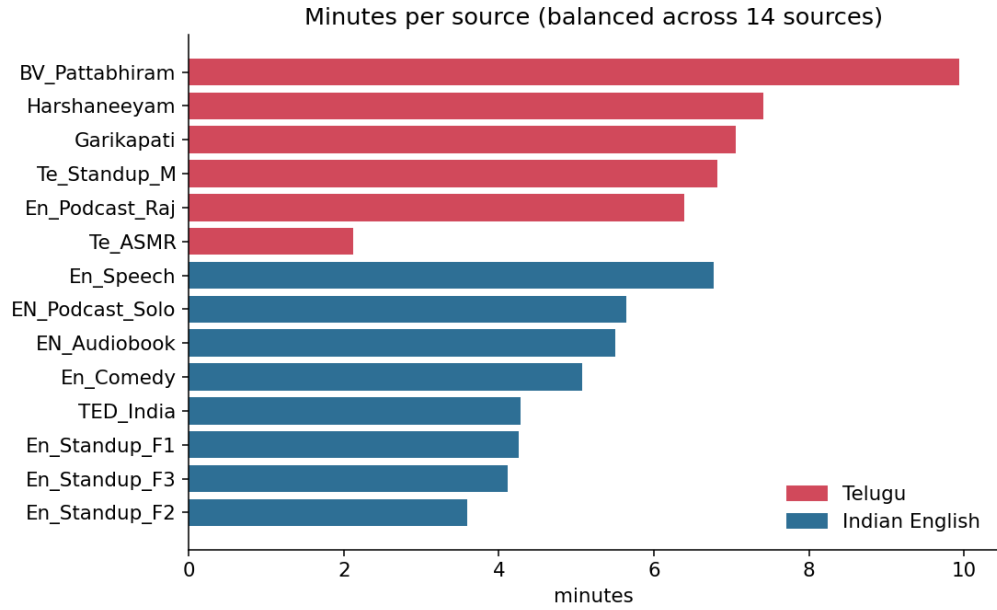


Figure 2: Minutes per source. The over-collect-then-balance step spreads the time evenly across all 14 sources, so no single channel dominates.

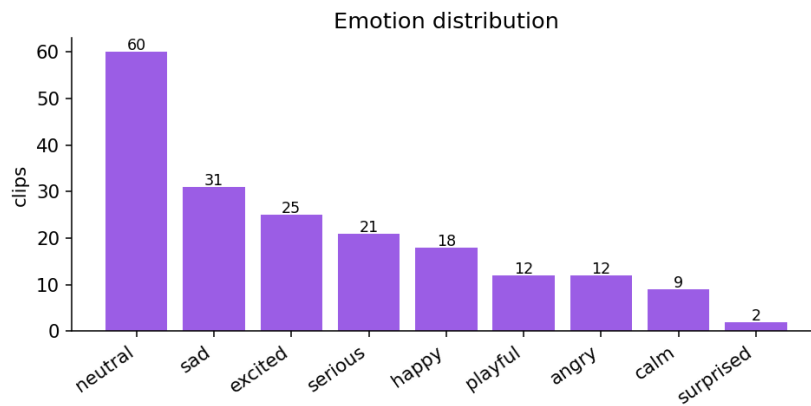


Figure 3: Emotion spread. Adding stand-up, podcasts and ASMR widened this from a neutral-only spike to nine emotions.

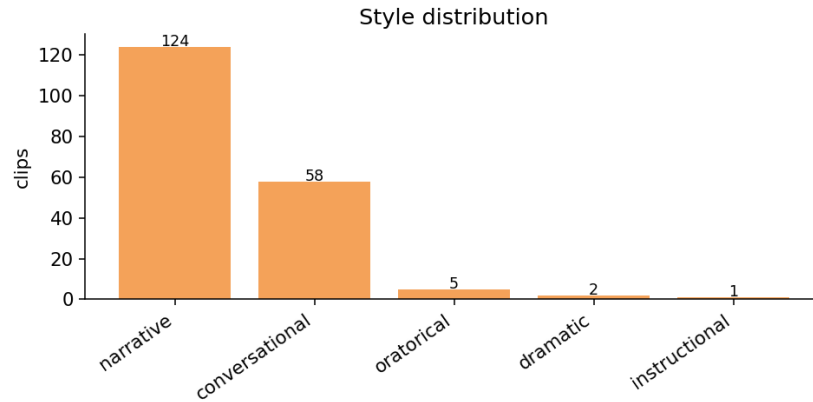


Figure 4: Speaking style. Still narrative-heavy, but conversational is now well represented.

- **Listening is the real work, so I made it easy.** I built a review app a non-technical person could use, and my Telugu-reading grandmother checked the Telugu clips – better ground truth than me, since I don’t read Telugu. She listened, confirmed or fixed the text, tapped an emoji for the feeling, and saved (or rejected bad clips). 37 verified, 13 rejected.

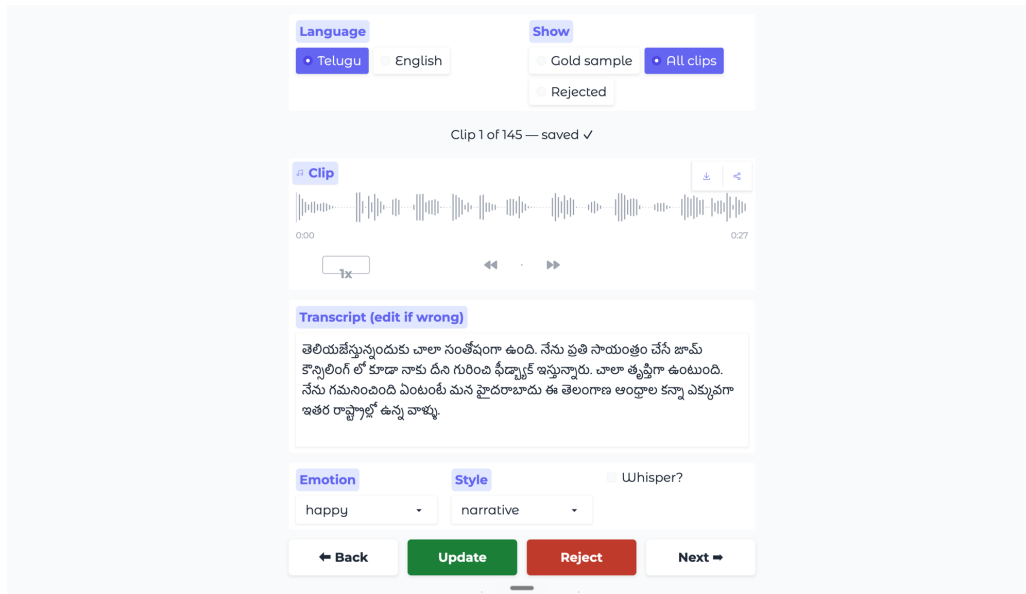


Figure 5: The review app: play the clip, fix the transcript if needed, tap how it sounds, Save or Reject. It opens on a small, balanced “review set” so you check a handful of important clips, not all 190.

## 5. What I’d improve given more time

**First, how I worked.** My biggest mistake was process, not code: once the first small test worked, I built the whole pipeline in one go and leaned heavily on AI agents to write it – so the silent bugs above only surfaced when I ran everything together. Next time I’d test each stage on two or three clips and check its output myself before moving to the next, and treat the AI as a pair-programmer





Figure 6: The bilingual version made for my grandmother – Telugu labels, big text, emoji for emotion so nothing needs reading in English, and a done/left counter.

I review rather than letting it drive.

### Then, the data:

- **Better noise detection** (the top fix). Instead of one blunt threshold, measure each clip's signal-to-noise ratio, add a model that specifically detects applause/laughter/crowd, gently denoise only the borderline clips, and send more of them to a human ear.
- **Shorter clips** (say 8-12 seconds). Smaller windows span fewer speaker turns, so there's far less chance of two voices landing in one clip – the main risk with podcasts.
- **Per-source silence settings** so quiet ASMR segments properly (this would recover the English ASMR I lost).
- **A multilingual emotion model** that actually knows Indian languages, so the audio and text emotion opinions agree more often.
- **More human review.** The 0.0% WER/CER is real but flattering – it reflects reviewers *accepting* Sarvam's transcripts verbatim, not a blind re-transcription. With more time I'd (a) review many more clips, and (b) have a few re-typed from scratch without seeing the ASR, so the number becomes a trustworthy *error* rate instead of an *approval* rate.
- **More female Telugu sources, more genres, and a YouTube scraper** to scale past a hand-picked roster.

## Ethics and licensing

Every clip records where it came from and what kind of source it is. I avoided copyrighted film and music, and dropped anything with a music bed. I haven't individually cleared redistribution rights for the underlying audio, so the dataset is meant for **research and educational use**; the licence tag covers the annotations I added. Anyone using it commercially should clear the source audio themselves.